GRAVITATION

Universal Central Fields, Planetary Metrics, and Keplerian Orbits

Subject: Classical Astrodynamics & Central Potential Fields

Level: Core Academic Reference Notes (Undergraduate Module)

Features: Full Vector Formulations & Native Orbital Geometries Embedded

Document Status: Final Production Master File

1. NEWTON'S UNIVERSAL LAW OF GRAVITATION

Gravitation is the universal attractive force acting between all material particles possessing mass properties. It acts as a long-range central force field that governs macroscopic cosmic configurations.

1.1 Mathematical Formulation & Vector Notation

Newton's Law states that the gravitational attraction force magnitude between any two point-mass objects (m_1 and m_2) is directly proportional to the product of their masses and inversely proportional to the square of the distance scalar r dividing their centers:

$$F = G \cdot (m_1 m_2) / r^2$$

Where G is the Universal Gravitational Constant value ($6.674 imes 10^{-11} ext{ N}\cdot ext{m}^2/ ext{kg}^2$). In rigorous vector coordinate fields, the force vector operating on mass 2 due to mass 1 is written as:

$$\&vec;F_{21} = - [G \cdot (m_1 m_2) / r^2] \cdot r_{12}$$

The negative sign mathematically indicates the inherently attractive central character of the force fields, operating opposite to the outwards radial tracking vector.

Figure 1.1: Direct Universal Vector Attraction alignment demonstrating Newton's action-reaction symmetry pairs along the central spatial line connecting m1 and m2.

2. ACCELERATION DUE TO GRAVITY (G) & FIELDANOMALIES

The gravitational field intensity on a planetary body creates a uniform localized acceleration parameter, denoted as g.

2.1 Base Formulation on Earth's Boundary Surface

Let M represent Earth's structural mass and R indicate its localized spatial radius profile. Equating standard point mass weight to the overarching central universal law yields:

$$m \cdot g = G \cdot (M \cdot m) / R^2 \quad g = G \cdot M / R^2$$

2.2 Variation of g with Altitude (Height h)

As a system moves an explicit vertical height threshold h above Earth's baseline boundary surface:

$$g_h = G \cdot M / (R + h)^2 = g \cdot [1 + h/R]^{-2}$$

Applying standard calculus binomial theorem expansions where we assume $h \ll R$ yields the linear drop profile:

$$g_h \approx g \cdot [1 - 2h/R]$$

2.3 Variation of g with Depth (d)

When tracking parameters inside a mine core structure at depth threshold d below the outer shell, only the interior mass sphere within radius $r = R - d$ exerts a net operational central field. This yields a direct proportional linear decay function:

Consequently, the localized acceleration due to gravity drops perfectly to zero at Earth's geometric center coordinate ($d = R$).

$$g_d = g \cdot [1 - d/R]$$

3. GRAVITATIONAL POTENTIAL ENERGY & ESCAPEVELOCITY METRICS

Because gravity acts as a conservative field system, we can analyze space tracking states using scalar potential field distributions.

3.1 Mathematical Derivation of Gravitational Potential Energy (U)

Potential energy is defined as the integrated negative mechanical work done to translate a target test mass from an absolute field-free infinity threshold ($r = \infty$) to a localized coordinate radius position r:

$$U(r) = - \int_{\infty}^{r} \&vec;F \cdot d\&vec;r = - \int_{\infty}^{r} [- GMm / x^{2}] \, dx$$

$$U(r) = GMm \cdot [-1/x]_{\infty}^{r} = - GMm / r$$

The absolute negative sign implies a bound field trap condition configuration—energy must be injected to separate the masses to infinity.

3.2 Derivation of Escape Velocity (v_e)

The **Escape Velocity** is the minimum launching speed required for a particle on a planetary surface to break completely free from its gravitational field trap, reaching infinity with zero residual kinetic energy.

Energy Conservation Method Proof: We establish total scalar mechanical energy balancing parameters across the launching surface and the infinity boundary limit:

Total Surface Energy: $E_{surface} = K_s + U_s = ½m(v_e)^2 - GMm/R$

Total Boundary Energy at Infinity: $E_{\infty} = K_{\infty} + U_{\infty} = 0 + 0 = 0$

Applying Total Energy Conservation principles ($E_{surface}$ = E_{\infty}):

$$½m(v_e)^2 - GMm/R = 0 \quad ½m(v_e)^2 = GMm/R$$

Canceling out the particle mass parameter m and extracting velocity vectors provides:

$$v_e = \sqrt{(2GM/R)} = \sqrt{(2gR)}$$

For Earth, substituting standard metrics yields an escape speed threshold of approximately **$11.2 ext{ km/s}$**.

4. SATELLITE KINEMATICS & ORBITAL DYNAMICS

Satellites in stable circular tracking paths balances inward gravitational central field attraction against outward coordinate kinematic requirements.

4.1 Orbital Velocity Function (v_o)

For a satellite of mass m orbiting at a radius tracking path $r = R + h$, the centripetal force parameter is supplied entirely by the localized universal attraction law:

$$m(v_o)^2 / r = GMm / r^2 \quad v_o = \sqrt{(GM/r)} = \sqrt{[GM / (R+h)]}$$

For low-altitude deployment zones ($h pprox 0$), the orbital profile simplifies to ***v_o = \sqrt{GM/R} = \sqrt{gR}***, yielding a baseline speed of **$7.92 ext{ km/s}$**.

4.2 Total Binding Energy profile of a Satellite

The total structural energy configuration of an active orbiting satellite combines its positive kinetic energy tracking state and negative potential position energy field location:

$$K = \tfrac{1}{2}m(v_o)^2 = GMm / 2r$$

$$U = - GMm / r$$

$$E_{total} = K + U = GMm/2r - GMm/r = - GMm / 2r$$

This negative result represents the satellite's **Binding Energy**. To extract the satellite from its orbit and send it into deep space, an equal amount of positive energy ($+GMm/2r$) must be added to the system.

5. KEPLER'S THREE LAWS OF PLANETARY MOTION

Johannes Kepler formulated three empirical laws detailing planetary tracking curves, which Sir Isaac Newton later validated mathematically using calculus principles.

5.1 The Law of Orbits (First Law)

All planets track paths that are explicitly **Elliptical** in geometry, with the central star hosting the system positioned at one of the two main focal coordinates.

5.2 The Law of Areas (Second Law)

A position line tracking vector drawn from the host star to a orbiting planet sweeps out equal geometric surface areas across equal intervals of time ($dA/dt = ext{constant}$). This is a direct mathematical expression of the **Conservation of Angular Momentum** ($ ec{L} = ext{constant}$) in central force fields.

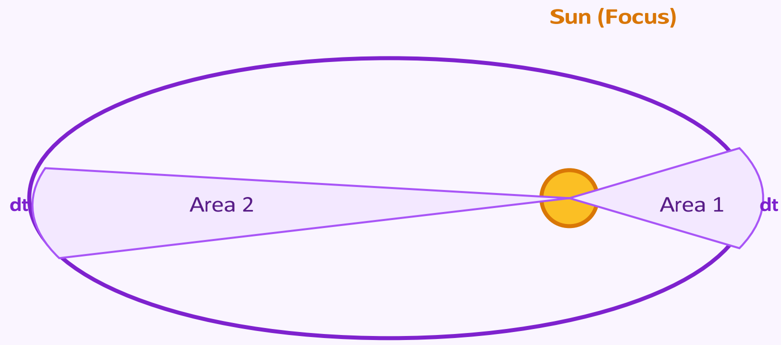

Figure 5.1: Kepler's Elliptical Orbit geometry demonstrating the Law of Areas. Area 1 matches Area 2 perfectly when tracked across equal time windows (dt).

5.3 The Law of Periods (Third Law

The square of the orbital period of revolution (***T***) of a planet is directly proportional to the cube of the semi-major axis parameter (***a***) of its elliptical tracking path:

$$T^2 \quad a^3$$

For a basic circular model where radius $\boldsymbol{r = a}$, this is easily proven by combining orbital velocity equations with standard circumference parameters:

$$T = 2\pi r / v_o = 2\pi r / \sqrt{(GM/r)} = 2\pi r^{3/2} / \sqrt{(GM)}$$

Squaring both sides $T^2 = [4\pi^2 / GM] \cdot r^3$